



**ADVANCED COMPUTER NETWORKS
(MTITE101)**

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Practical 2:

1. Aim: To capture and analyze IPv4 and IPv6 packets using Wireshark and study their respective protocol header fields.

2. Commands Run

IPv4:

`ping -c 4 8.8.8.8`

Function: Generates ICMP-based IPv4 traffic for packet capture and analysis in Wireshark.

Output:

No.	Time	Source	Destination	Protocol	Length	Info
453	72.501714	192.168.1.2	8.8.8.8	ICMP	98	Echo (ping) request id=0x0282, seq=0/0, ttl=64 (reply in 454)
454	72.513140	8.8.8.8	192.168.1.2	ICMP	98	Echo (ping) reply id=0x0282, seq=0/0, ttl=118 (request in 453)
459	73.505825	192.168.1.2	8.8.8.8	ICMP	98	Echo (ping) request id=0x0282, seq=1/256, ttl=64 (reply in 460)
460	73.518294	8.8.8.8	192.168.1.2	ICMP	98	Echo (ping) reply id=0x0282, seq=1/256, ttl=118 (request in 459)
464	74.511067	192.168.1.2	8.8.8.8	ICMP	98	Echo (ping) request id=0x0282, seq=2/512, ttl=64 (reply in 465)
465	74.520894	8.8.8.8	192.168.1.2	ICMP	98	Echo (ping) reply id=0x0282, seq=2/512, ttl=118 (request in 464)
466	75.516269	192.168.1.2	8.8.8.8	ICMP	98	Echo (ping) request id=0x0282, seq=3/768, ttl=64 (reply in 467)
467	75.526632	8.8.8.8	192.168.1.2	ICMP	98	Echo (ping) reply id=0x0282, seq=3/768, ttl=118 (request in 466)

> Frame 453: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface	0000	14 33 75 ed e7 20 fa da 32 05 de 58 08 00 45 00
> Ethernet II, Src: fa:da:32:05:de:58 (fa:da:32:05:de:58), Dst: ZyxeICommuni_ed:e7:20 (14:33	0010	00 54 41 c4 00 00 40 01 67 2b c0 a8 01 02 08 08
> Internet Protocol Version 4, Src: 192.168.1.2, Dst: 8.8.8.8	0020	08 08 08 00 3d 67 02 82 00 00 6a 78 8b ed 00 03
> 0100 = Version: 4	0030	d6 aa 08 09 0a 0b 0c 0d 0e 0f 10 11 12 13 14 15
> 0101 = Header Length: 20 bytes (5)	0040	16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)	0050	26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35
> Total Length: 84	0060	36 37
> Identification: 0x41c4 (16836)		
> 000. = Flags: 0x0		
> ...0 0000 0000 0000 = Fragment Offset: 0		
> Time to Live: 64		
> Protocol: ICMP (1)		
> Header Checksum: 0x672b [validation disabled]		
> [Header checksum status: Unverified]		
> Source Address: 192.168.1.2		
> Destination Address: 8.8.8.8		
> [Stream index: 5]		
> Internet Control Message Protocol		

IPv6:

ping6 -c 4 ::1

Function: Generates ICMPv6 traffic using the IPv6 loopback address for packet capture and analysis in Wireshark.

Output:

The image displays a Wireshark packet capture of ICMPv6 traffic. The packet list on the left shows several Neighbor Solicitation (NS) and Router Advertisement (RA) messages. The packet details pane on the right shows the structure of an ICMPv6 Neighbor Solicitation message.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
3	1.331401	fe80::1633:75ff:fe...	2401:4900:8843:716...	ICMPv6	86	Neighbor Solicitation for 2401:4900:8843:716c:788f:cbd4:524d:12aa from 14:33:75:ed:e7:20
4	1.331542	fe80::f3:b653:aa03...	fe80::1633:75ff:fe...	ICMPv6	78	Neighbor Advertisement 2401:4900:8843:716c:788f:cbd4:524d:12aa (sol)
6	1.816752	fe80::1633:75ff:fe...	ff02::1	ICMPv6	174	Router Advertisement from 14:33:75:ed:e7:20
12	3.192282	2401:4900:8843:716...	2403:300:1364::2:1	ICMPv6	131	Destination Unreachable (Port unreachable)
180	27.728444	fe80::f3:b653:aa03...	fe80::1633:75ff:fe...	ICMPv6	86	Neighbor Solicitation for fe80::1633:75ff:feed:e720 from fa:da:32:05:de:58
181	27.734930	fe80::1633:75ff:fe...	fe80::f3:b653:aa03...	ICMPv6	78	Neighbor Advertisement fe80::1633:75ff:feed:e720 (rtr, sol)
195	32.807584	fe80::1633:75ff:fe...	fe80::f3:b653:aa03...	ICMPv6	86	Neighbor Solicitation for fe80::f3:b653:aa03:8cac from 14:33:75:ed:e7:20
196	32.807672	fe80::f3:b653:aa03...	fe80::1633:75ff:fe...	ICMPv6	78	Neighbor Advertisement fe80::f3:b653:aa03:8cac (sol)
200	41.881881	fe80::1633:75ff:fe...	ff02::1	ICMPv6	174	Router Advertisement from 14:33:75:ed:e7:20
325	58.368323	fe80::1633:75ff:fe...	2401:4900:8843:716...	ICMPv6	86	Neighbor Solicitation for 2401:4900:8843:716c:788f:cbd4:524d:12aa from 14:33:75:ed:e7:20
326	58.368472	fe80::f3:b653:aa03...	fe80::1633:75ff:fe...	ICMPv6	78	Neighbor Advertisement 2401:4900:8843:716c:788f:cbd4:524d:12aa (sol)
378	72.486094	fe80::f3:b653:aa03...	fe80::1633:75ff:fe...	ICMPv6	86	Neighbor Solicitation for fe80::1633:75ff:feed:e720 from fa:da:32:05:de:58
380	72.492817	fe80::1633:75ff:fe...	fe80::f3:b653:aa03...	ICMPv6	78	Neighbor Advertisement fe80::1633:75ff:feed:e720 (rtr, sol)
418	77.620258	fe80::1633:75ff:fe...	fe80::f3:b653:aa03...	ICMPv6	86	Neighbor Solicitation for fe80::f3:b653:aa03:8cac from 14:33:75:ed:e7:20
419	77.620397	fe80::f3:b653:aa03...	fe80::1633:75ff:fe...	ICMPv6	78	Neighbor Advertisement fe80::f3:b653:aa03:8cac (sol)
420	81.926682	fe80::1633:75ff:fe...	ff02::1	ICMPv6	174	Router Advertisement from 14:33:75:ed:e7:20
532	93.799669	fe80::1633:75ff:fe...	2401:4900:8843:716...	ICMPv6	86	Neighbor Solicitation for 2401:4900:8843:716c:788f:cbd4:524d:12aa from 14:33:75:ed:e7:20
534	93.799813	fe80::f3:b653:aa03...	fe80::1633:75ff:fe...	ICMPv6	78	Neighbor Advertisement 2401:4900:8843:716c:788f:cbd4:524d:12aa (sol)
597	114.575621	fe80::f3:b653:aa03...	fe80::1633:75ff:fe...	ICMPv6	86	Neighbor Solicitation for fe80::1633:75ff:feed:e720 from fa:da:32:05:de:58
588	114.582394	fe80::1633:75ff:fe...	fe80::f3:b653:aa03...	ICMPv6	78	Neighbor Advertisement fe80::1633:75ff:feed:e720 (rtr, sol)
621	119.706881	fe80::1633:75ff:fe...	fe80::f3:b653:aa03...	ICMPv6	86	Neighbor Solicitation for fe80::f3:b653:aa03:8cac from 14:33:75:ed:e7:20
622	119.707825	fe80::f3:b653:aa03...	fe80::1633:75ff:fe...	ICMPv6	78	Neighbor Advertisement fe80::f3:b653:aa03:8cac (sol)

Packet Details:

Frame 3: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface en
> Ethernet II, Src: ZyxelCommuni_ed:e7:20 (14:33:75:ed:e7:20), Dst: fa:da:32:05:de:58 (fa:da:32:05:de:58)
> Internet Protocol Version 6, Src: fe80::1633:75ff:feed:e720, Dst: 2401:4900:8843:716c:788f:cbd4:524d:12aa
0110 = Version: 6
> 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
> 0000 0000 0000 0000 0000 = Flow Label: 0x00000
Payload Length: 32
Next Header: ICMPv6 (58)
Hop Limit: 255
> Source Address: fe80::1633:75ff:feed:e720
> Destination Address: 2401:4900:8843:716c:788f:cbd4:524d:12aa
[Source SLAAC MAC: ZyxelCommuni_ed:e7:20 (14:33:75:ed:e7:20)]
[Stream index: 1]
> Internet Control Message Protocol v6
Type: Neighbor Solicitation (135)
Code: 0
Checksum: 0x7586 [correct]
[Checksum Status: Good]
Reserved: 00000000
Target Address: 2401:4900:8843:716c:788f:cbd4:524d:12aa
> ICMPv6 Option (Source Link-Layer address : 14:33:75:ed:e7:20)

3. Result: IPv4 and IPv6 packets were successfully captured and analyzed using Wireshark. The corresponding IPv4 and IPv6 header fields were observed and studied.